

22.To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *numbers;

    numbers = malloc(5 * sizeof(int));

    if (numbers == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    for (int i = 0; i < 5; i++)
        numbers[i] = i + 1;

    printf("Numbers:\n");

    for (int i = 0; i < 5; i++)
        printf("%d ", numbers[i]);

    printf("\n");

    free(numbers);

    printf("Memory released successfully.\n");

    return 0;
}
```

```
bindu@Bindusree:~$ nano skill22.c
bindu@Bindusree:~$ gcc skill22.c -o skill22
bindu@Bindusree:~$ ./skill22
Numbers:
1 2 3 4 5
Memory released successfully.
bindu@Bindusree:~$ |
```